

THE BLUE BRAIN PROJECT: Unraveling The Brain's Mystery



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The human brain is the most remarkable creation of God and also the reason behind human intelligence. Blue Brain is the name of the world's first virtual brain—a machine that works on the same lines as human brain. The Blue Brain System is an endeavour to reverse-engineer the human brain and rebuild it at the molecular level via computer simulation. The project was initiated in May 2005 by Prof. Henry Markram at Ecole Polytechnique Fédérale de Lausanne (EPFL) in Lausanne, Switzerland.

IBM (fondly known as The Big Blue) and Swiss University team, associated with fabricating a custom-made super-computer based on IBM's Blue Gene design, have been working together on Blue Brain Project. IBM brings to the table expertise in visualisation, simulations, algorithms, Blue Gene optimisations and development of innovative computational methods.

Blue Brain Project expects researchers to develop their own models of various brain areas in dissimilar species and at altered levels of detail using Blue Brain Software for simulation on Blue Gene. The objective is to collect these models in a central Internet database from which Blue Brain Software can mine and link models together to form brain regions. This would finally lead to the first whole brain simulation.

It is hoped that Blue Brain Project will enable

building up the virtual brain, which will ultimately unravel the mysteries of the key facets of human cognition, such as perception, memory and, may be, consciousness, too. For the first time, we will be able to witness the electrical code our brains use to denote the world, in real-time basis. We also expect to gain an understanding of how certain failures of the brain's microcircuits lead to psychiatric disorders such as autism, schizophrenia and depression.

Reconstructing the enigmatic human brain using IT

Typically, a human brain has over 100 billion neurons or brain cells and about 100 trillion synapses, thus making it a very complex multilevel system. The inter-neuron connections make up a hierarchy of circuits that range from local microcircuits to macro-circuits that form the entire brain. At the unit level, each neuron and synapse is a complex molecular machine in itself. It is the interactions between these levels that leads to human behaviour, human emotion and human cognition, as we know it.

Blue Brain Project aspires to develop wide-ranging digital reconstructions, that is, computer models of the brain including its diverse levels of organisation and interactions. Blue Brain Project's reconstruction approach pinpoints interdependencies in experimental data; for example, dependencies between the size of neurons and neuron densities, dependencies between the shapes of neurons and the synapses these form, and dependencies between the number of boutons on axons and synapse numbers. It then uses these to coerce the reconstruction procedure. Multiple and intersecting constraints enable the project to develop the most accurate reconstructions possible from the scanty experimental data available, thus



Fig. 1: Professor Henry Markram, father of Blue Brain Project (Image courtesy: Frankenstein Project)



Fig. 2: Blue Brain is the name of the world's first virtual brain (Image courtesy: www.alyellowpage.com)